

CLD Studio

User Guide

⚠ This tool is under active development. Some calculations or displays may still show errors or unhandled cases (the "LOG" panel, at the bottom right, reports any calculation anomalies detected). On the plus side, the tool is easy to modify: any behavior, calculation, or feature can be adjusted or corrected quickly on request.

CLD Studio is a tool for building and analyzing Causal Loop Diagrams. It lets you model a system as a set of variables linked by cause-and-effect relationships, automatically detect feedback loops (reinforcing or balancing), and simulate their behavior over time.

This guide uses a classic teaching example — the dynamics of a population based on its birth rate and death rate — to introduce the tool's general features. It then draws on a real timber-construction project to illustrate the more advanced features (Data tables, 3D Model).

This timber-construction project focuses specifically on Timber Frame Facades (TFF): this is what motivated the development of the 3D Model, designed to represent and calculate a cross-section of a TFF panel from the diagram's variables.

1. Overview

On first launch, the screen is empty and prompts you to create a variable or import an existing Excel file (File menu).

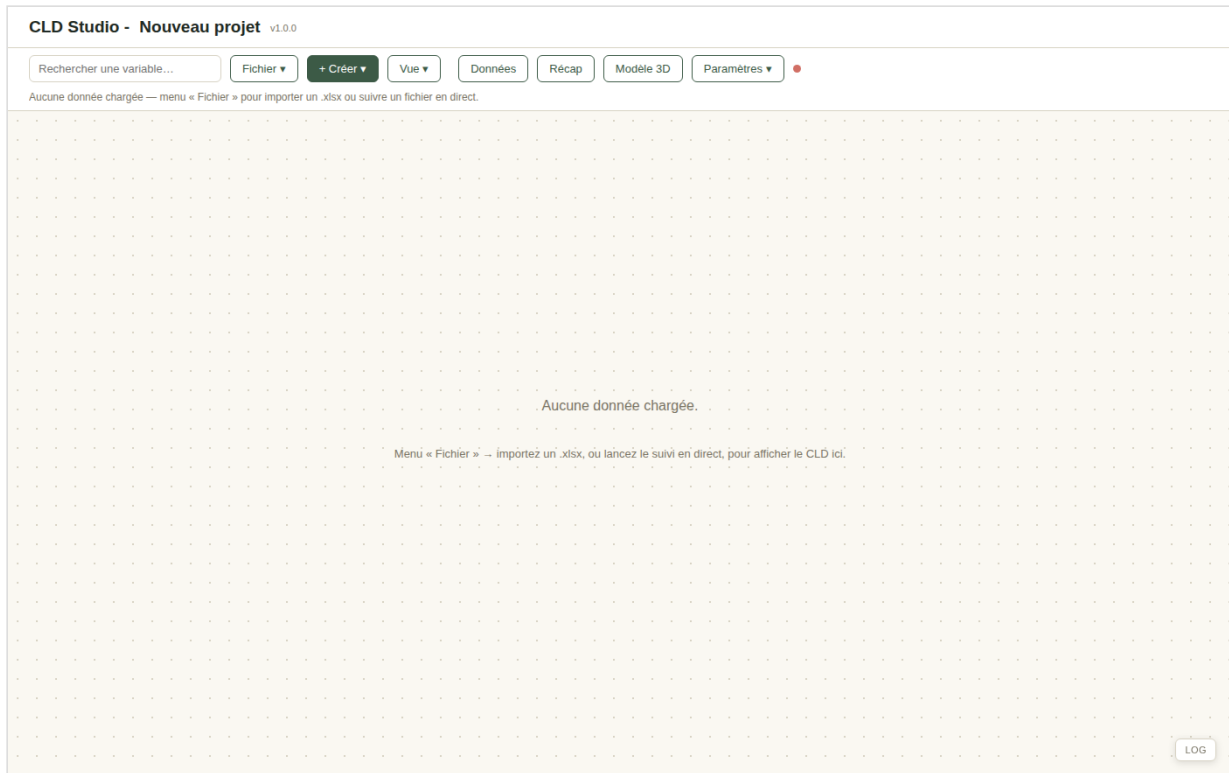


Figure 1. Startup screen, with no data loaded.

The toolbar, always visible at the top of the screen, gives access to all the features:

- **File:** import/export a project, follow an Excel file live, export the graph as a PDF or image.
- **+ Create:** add a new variable or a new causal link.
- **View:** see the list of detected loops, reorganize the graph by category, filter the display, view the tag legend.
- **Data:** manage reference data tables (e.g., a materials catalog) usable in formulas.
- **Summary:** an editable overview of all variables, grouped by the origin of their value.
- **3D Model:** for timber-construction projects, an interactive 3D model driven by the diagram's variables.
- **Settings:** interface language, background variable transparency, startup guide.

The tool's version number is displayed permanently next to the project name, at the top of the screen.

2. Creating variables

A variable represents a quantity in the system being studied: an amount, a rate, an indicator, etc. It is created from the "+ Create" menu → "+ New variable" (keyboard shortcut: V).

CLD Studio - Nouveau projet v1.0.0

Rechercher une variable...

Fichier

+ Créer

Vue

Données

Récap

Modèle 3D

Paramètres

Aucune donnée chargée — menu « Fichier » pour importer un .xlsx ou suivre un fichier en direct.

Aucune donnée chargée.

Menu « Fichier » → Importez un .xlsx, ou lancez le suivi en direct, pour afficher le CLD ici.

Retour aux boucles

Nouvelle variable

NOM *

ex: Qualité du bois source

UNITE

ex: kg, %, €/MOB...

VALEUR (HYPOTHESE)

ex: 10

FORCE

Moyenne

TAG / CATEGORIE

optionnel

DESCRIPTION

optionnel

Créer la variable

LOG

Figure 2. Variable creation form: name, unit, hypothesis value, strength, tag.

For our example, we create three variables: "Population" (1000 inhabitants), "Birth rate" (2%), and "Death rate" (1.5%). Each variable can be freely moved on the graph by drag-and-drop.

CLD Studio - Nouveau projet v1.0.0

Rechercher une variable...

Fichier

+ Créer

Vue

Données

Récap

Modèle 3D

Paramètres

Variable V3 créée : Taux de mortalité

Taux de natalité

Population

Taux de mortalité

V3

NOM DE LA VARIABLE

Taux de mortalité

UNITE

%

VALEUR RETENUE

1.5 % (hypothèse)

INCERTITUDE (±)

ex: ± 10 %

VALEUR (HYPOTHESE)

1.5

☐ Calculer via une formule
 ☐ Limites

FORCE DE LA VARIABLE

Moyenne

TAG / CATEGORIE

ex: Dangersité de la mise en oeuvre

CONNEXIONS (0)

DESCRIPTION

Ajouter une description...

Supprimer cette variable

LOG

Figure 3. The three example variables, positioned on the graph.

3. Creating causal links

A link connects two variables and carries a polarity:

- + (same direction): when the source increases, the target increases too (all else being equal).
- – (opposite direction): when the source increases, the target decreases.
- ?: the relationship exists but its direction has not yet been established (useful in the draft stage).

We connect our three variables as follows:

- Population → Birth rate (+): more population, more births in absolute value.
- Birth rate → Population (+): births increase the population.
- Population → Death rate (+): more population, more deaths in absolute value.
- Death rate → Population (–): deaths decrease the population.

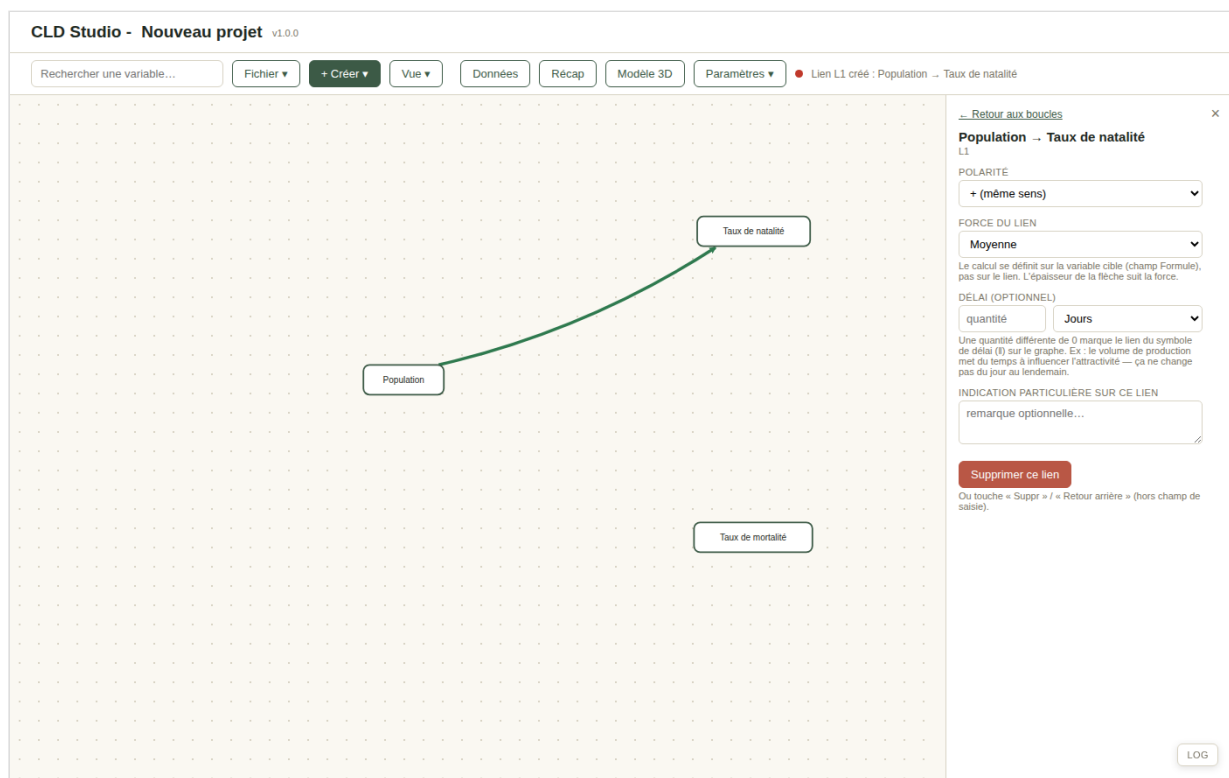


Figure 4. Link creation form: source, target, polarity, strength, delay.

Once the four links have been created, the complete graph shows two green arrows ("+" relationships) and one red arrow ("–" relationship).

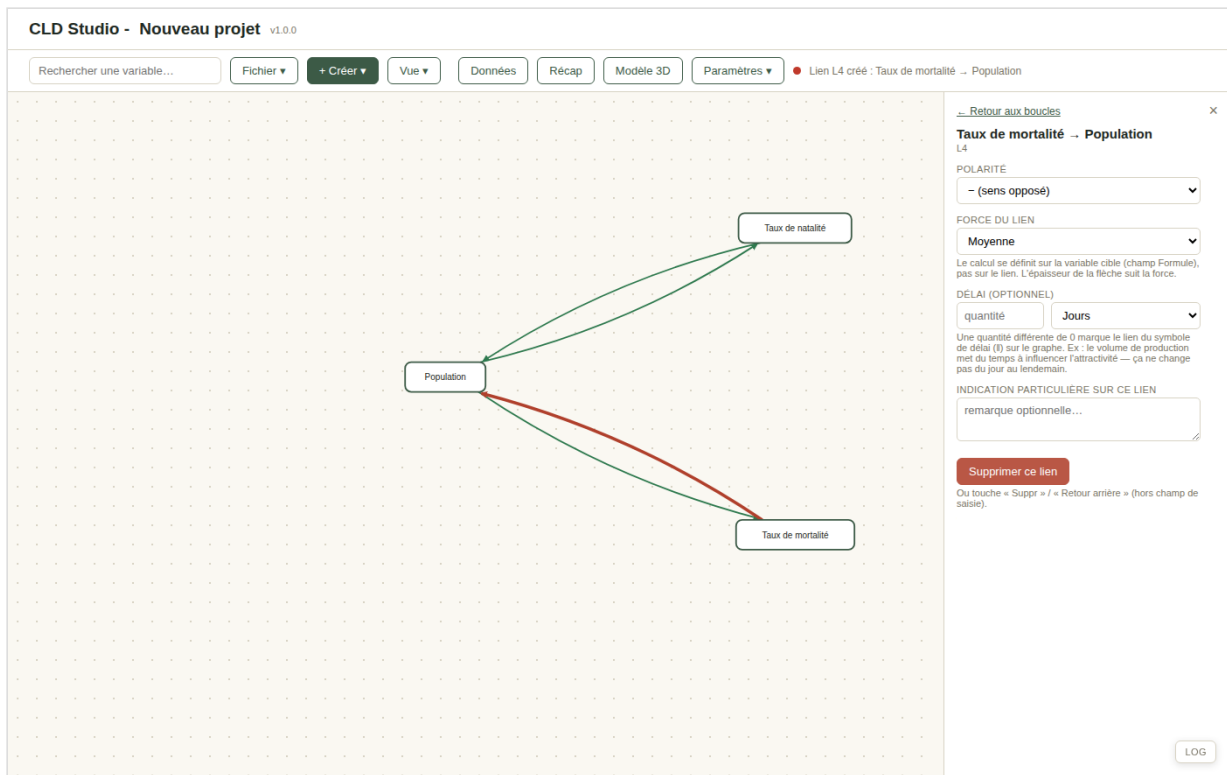


Figure 5. The complete graph: Population, Birth rate and Death rate, connected by four causal links.

Clicking on a variable opens its detail panel: name, unit, current value, hypothesis, strength, tag, and the list of its incoming and outgoing connections.

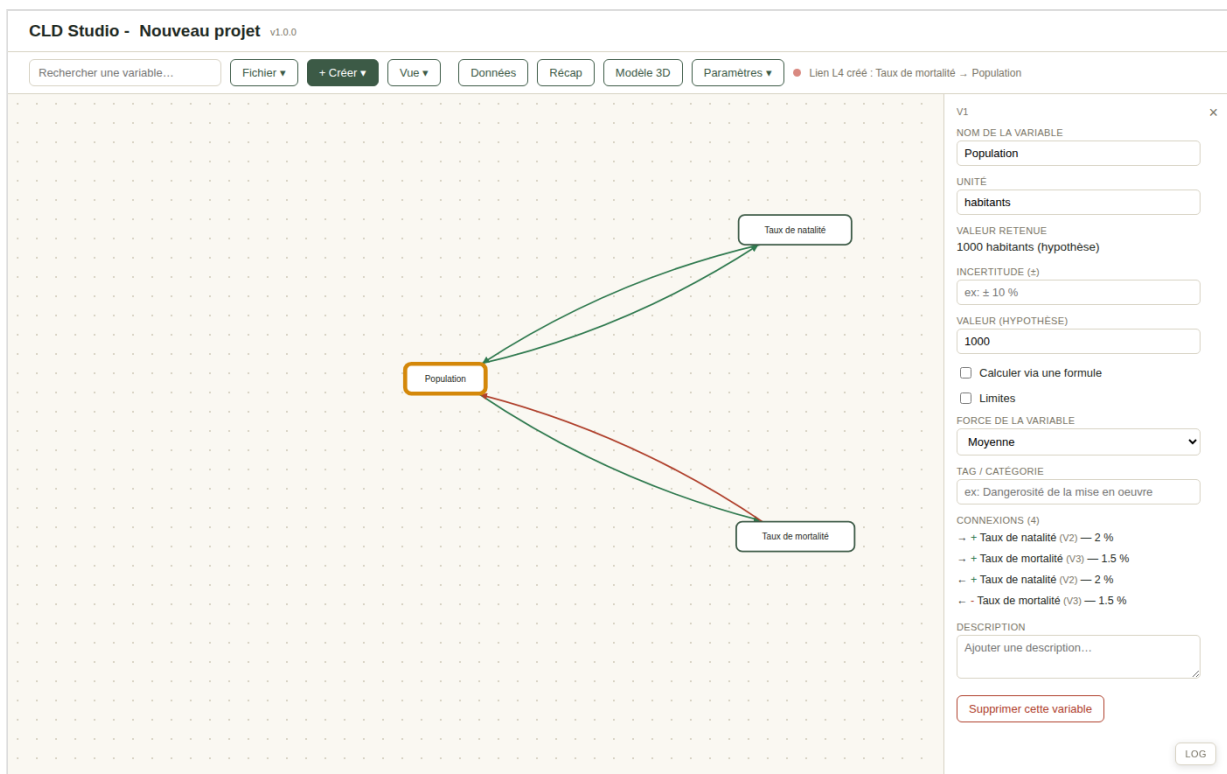


Figure 6. Detail panel for the "Population" variable, with its four connections.

4. Detecting and analyzing loops

This is the heart of the tool: as soon as the graph contains a cycle, CLD Studio detects it automatically and displays it under "View → View loops", along with its type:

- **R (reinforcing)**: an even number of "-" links in the cycle: the effect self-amplifies (growth or exponential collapse).
- **B (balancing)**: an odd number of "-" links: the effect stabilizes around an equilibrium.

Our example produces exactly these two cases:

- Population → Birth rate → Population: two "+" links → loop R. The more inhabitants there are, the more births there are, which further increases the population: a self-sustaining growth dynamic.
- Population → Death rate → Population: one "+" link and one "-" link → loop B. The more inhabitants there are, the more deaths there are, which slows the population's growth: a dynamic that tends toward equilibrium.

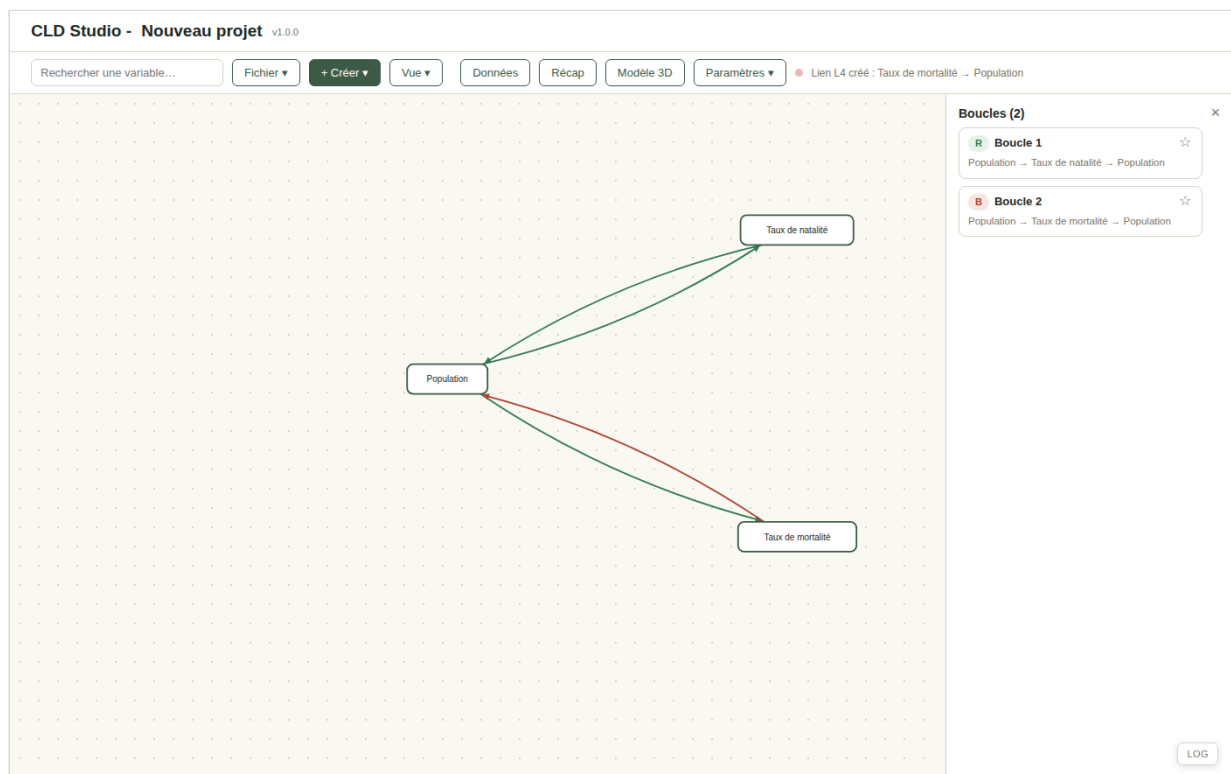


Figure 7. The two loops detected automatically: R (growth) and B (mortality).

A loop can be marked as a favorite (star icon) so it can be found quickly in a complex diagram with many loops. Opening a loop also lets you display its neighboring variables at proximity levels (1, 2, or 3 hops), to explore its context without being overwhelmed by the whole diagram. A "Reorganize graph" checkbox lets you choose whether increasing the proximity temporarily rearranges the loop and its neighbors for better readability, or whether variables stay in their usual position on the graph.

Selecting a variable or a loop actually grays out the rest of the graph (not just a simple transparency effect): gray fill, darker outline to stay legible, and text that remains black and crisp. A legend (link polarity, proximity level, delay, tags) appears in the View → Tags menu, and is automatically included in PDF and image exports. This graying-out can be disabled in Filters ("Gray out unselected variables/links") to keep the whole graph at full color even during a selection.

5. Calculating and iterating

The "Summary" panel groups all the project's variables, classified by the origin of their value (hypothesis, data-derived, formula-calculated, or with no value), and lets you run an iterative calculation: at each iteration, the tool recalculates every variable that has a formula or a linked data value, taking into account any delays on the links.

Récapitulatif des paramètres Toutes les variables, regroupées selon l'origine de leur valeur, triées par force. ☒ Faible ☒ Moyenne ☒ Élevée

Itérations Ligne Délai (s)

1. Hypothèses (modifiables) (3)

VARIABLE	FORCE	UNITÉ	VALEUR (HYPOTHÈSE)
Population (V1)	Moyenne	habitants	1000
Taux de mortalité (V3)	Moyenne	%	1.5
Taux de natalité (V2)	Moyenne	%	2

2. Valeur issue de Données (0)
Aucune variable dans cette catégorie.

3. Valeur = formule (0)
Aucune variable dans cette catégorie.

4. Sans valeur ni formule (0)
Aucune variable dans cette catégorie.

Figure 8. The Summary panel, with our three editable hypotheses.

In our example, none of the three variables has a formula: they are all starting hypotheses. To observe the population's dynamics over time, "Population" would need a formula such as "previous value + (births – deaths)", which follows exactly the same logic as the timber-construction example detailed further below, which makes full use of formulas and loops.

6. Filtering and organizing the view

On a diagram with several dozen variables, the filters (View → Filters menu) let you bring a subset to the front without removing the rest of the context: variables that are not concerned are simply grayed out, not hidden.

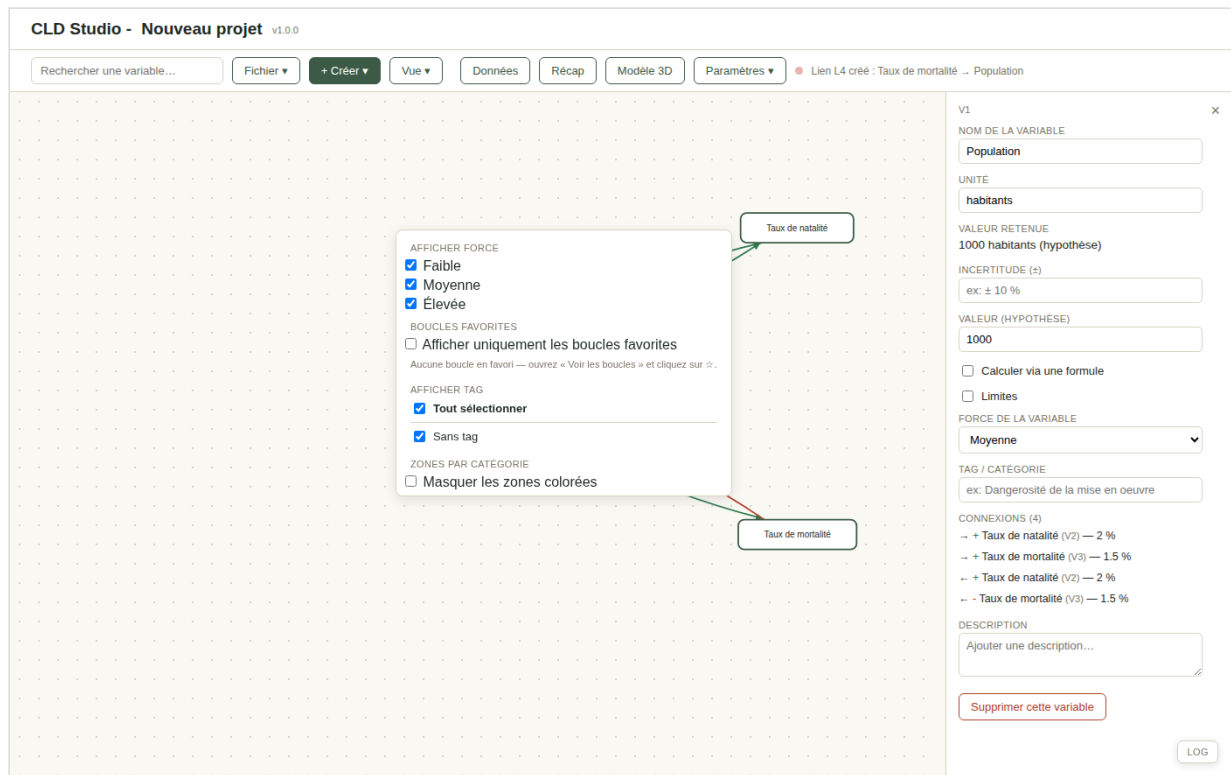


Figure 9. The Filters panel: strength, favorite loops, tags, colored zones.

- **Show strength:** grays out variables of Low, Medium, or High strength.
- **Favorite loops:** only clearly displays the variables of loops marked as favorites.
- **Show tags:** grays out variables according to their category.
- **Hide colored zones:** in "By category" view, hides the colored territories without changing the layout of the variables.

Two independent sliders, in Settings, adjust the gray tint applied to unselected variables and links respectively (the higher the percentage, the darker and more visible the gray); this setting also applies to PDF/image exports.

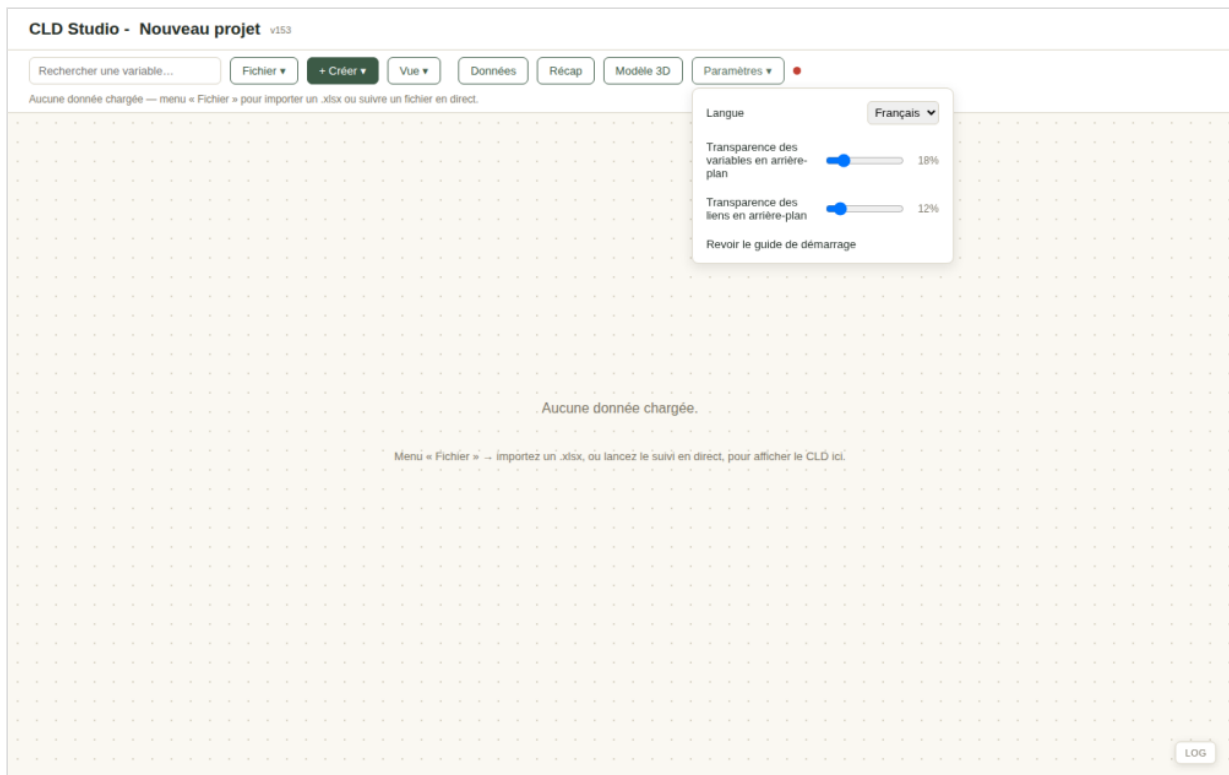


Figure 10. The Settings menu: language, background variable transparency, startup guide.

7. Importing, exporting, sharing

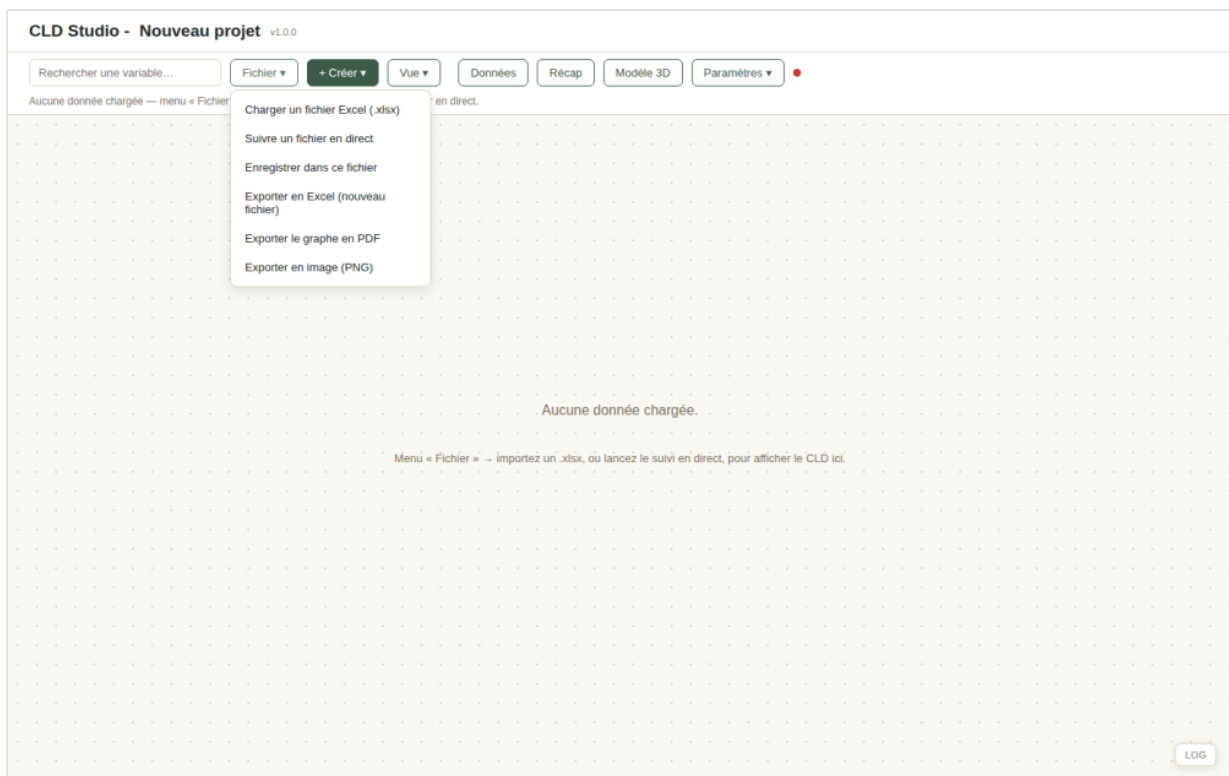


Figure 11. The File menu.

- **Load an Excel file (.xlsx):** reopens a previously exported project.

- **Follow a file live:** connects to an Excel file on disk: any change to it is automatically reloaded into the tool.
- **Save to this file:** writes directly to the file being followed live (shortcut Ctrl+S).
- **Export as Excel:** downloads a new .xlsx file with the entire project (variables, links, data, functions, 3D model configuration).
- **Export graph as PDF:** a PDF image of the diagram, laid out on a single page.
- **Export as image (PNG):** a crisp image of the diagram at its actual size, with no imposed page layout — preferable when the PDF crops content (for example with extended colored zones).

The exported Excel file is structured into several tabs (Variables, Links, Data, Functions, 3D Model, Project) that encode the entire project. A project can thus be shared, re-imported, or edited directly in Excel before being re-imported.

PDF and image exports now automatically include a legend (link polarity, proximity level, delay, tags) directly in the graph, and the PDF's title includes the project name and version number shown in the application, so it stays identifiable once downloaded.

8. Data tables

The "Data" button opens reference tables, for example a materials catalog, in which each column can be linked to a CLD variable, or used as a source for a Function. This is the bridge between a real product catalog (insulation, cladding, etc.) and the causal diagram.

Données — isolants
+ Ligne
+ Colonne
Fermer

Ligne	Type d'isolant	Forme de l'isolant	Prix unitaire isolant	Epaisseur d'isolant/ossature	Résistance thermique de l'isolant	Impact carbone de l'isolant	Euroclasse de l'isolant	Densité de l'isolant	coût du FOB moyen/Requiv/m²	Capacité thermique massique de l'isolant
GR 32 R	Laine de ver	Panneau	17.22	45	1.4	1.67	7	25	85.77	900
GR 32 R	Laine de ver	Panneau	30.25	140	4.35	4.34	7	25	31.92	900
GR 32 R	Laine de ver	Rouleau	16.51	100	3.15	3.42	7	25	38.98	900
GR 32 R	Laine de ver	Rouleau	26.21	160	5	5.18	7	25	27.26	900
Isolation	Paille	Botte	20	220	4.6	3.59	3	105	29.22	1558
Isolation	Paille	Botte	25	360	7.5	6.07	3	105	19.84	1558
Isolation	Paille	hachée	28	221	4.8	3.36	6	110	29.63	1558
Isolation	Paille	hachée	50	400	8.69	6.12	6	110	20.29	1558

Fonctions du tableau « Isolants »
+ Ajouter une fonction

Nouvelle fonction

TABLEAU SOURCE: Isolants
VARIABLE SOURCE (X): coût du FOB moyen/Requiv/m²

VARIABLE CIBLE (Y) — DANS LE CLD: Indice coût de revient FOB/ Attractivité (V135)

Y QUAND X = MIN: 0
Y QUAND X = MOYENNE (OPTIONNEL): 1
Y QUAND X = MAX: 2

$$f(x) = 0.0694 \cdot x - 1.3777 \text{ si } x \leq \text{moyenne (34.241111)}, \text{ sinon } 0.0194 \cdot x + 0.3355 \text{ (x = coût du FOB moyen/Requiv/m}^2\text{, borné entre 19.84 et 85.77)}$$

La moyenne de x (la colonne source) est calculée automatiquement. Le champ « y quand x = moyenne » ci-dessus est optionnel : laissez-le vide pour une simple droite proportionnelle entre min et max (ex: 0 à 2) ; renseignez-le uniquement si vous voulez forcer un point de passage précis à la moyenne (la courbe devient alors une ligne brisée en deux segments).
Variable cible : la variable du CLD qui recevra automatiquement cette valeur calculée (à la place de sa valeur manuelle) lors du calcul itératif, pour la ligne sélectionnée dans le menu « Itératif ».

LOG

Figure 12. An "Insulation" table: each row is a product, and each column can be linked to a variable (the  icon); below, a Function $f(x)$ computes a target variable from a source column.

- **Multiple tables:** accessible via tabs (here: Insulation, Insulation data, Interior bracing, Exterior bracing); "+ Table" creates a new one.
- **Linking a column to a variable:** the  icon links a column to an existing CLD variable: selecting a row in the table then updates that variable in the calculation.
- **Functions:** instead of a direct link, a Function computes a target variable from a source column by interpolation (min / average / max), for relationships that are not strictly proportional.
- **+ Row / + Column:** extend the table; a column can itself be computed by a formula (e.g., a cost per square meter derived from a price and an area).

9. 3D Model

For timber-construction projects, the "3D Model" button opens an interactive 3D cross-section of the panel (framing, insulation, cladding), driven by the project's variables and data tables rather than drawn by hand.

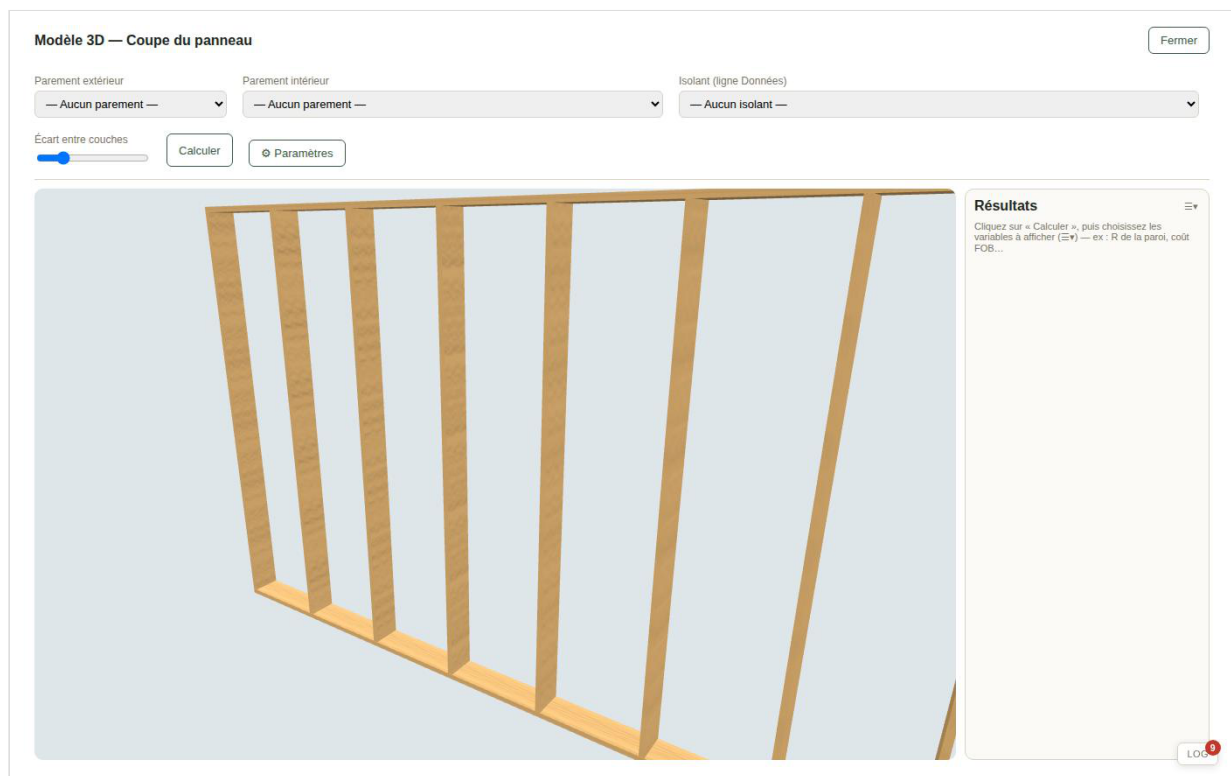


Figure 13. General view: timber frame with studs, before selecting an insulation or a cladding.

- **Exterior / interior cladding:** choose a row in a linked Data table (e.g., Exterior/Interior bracing); the appearance and thickness of the layer follow the selected row.
- **Insulation (Data row):** select a row from the main table; its thickness, texture, and properties drive the insulation layer.

- **Layer spacing:** a slider to visually separate the layers in the cross-section view, without changing their order.

The "⚙ Settings" button links each dimension of the model (panel width and height, insulation/framing thickness, stud width and spacing) to a CLD variable, with search and a preview of the current value:

Figure 14. 3D model settings: each dimension is linked to a variable (with its current value shown below), and each cladding to its Data table.

The "Calculate" button runs the full iterative calculation for the project (10 iterations) based on the selections made in this panel. The "Results" panel, filterable like the iterations table, then shows the variables of its choice (thermal resistance, cost, mass, phase shift, etc.) recalculated accordingly:

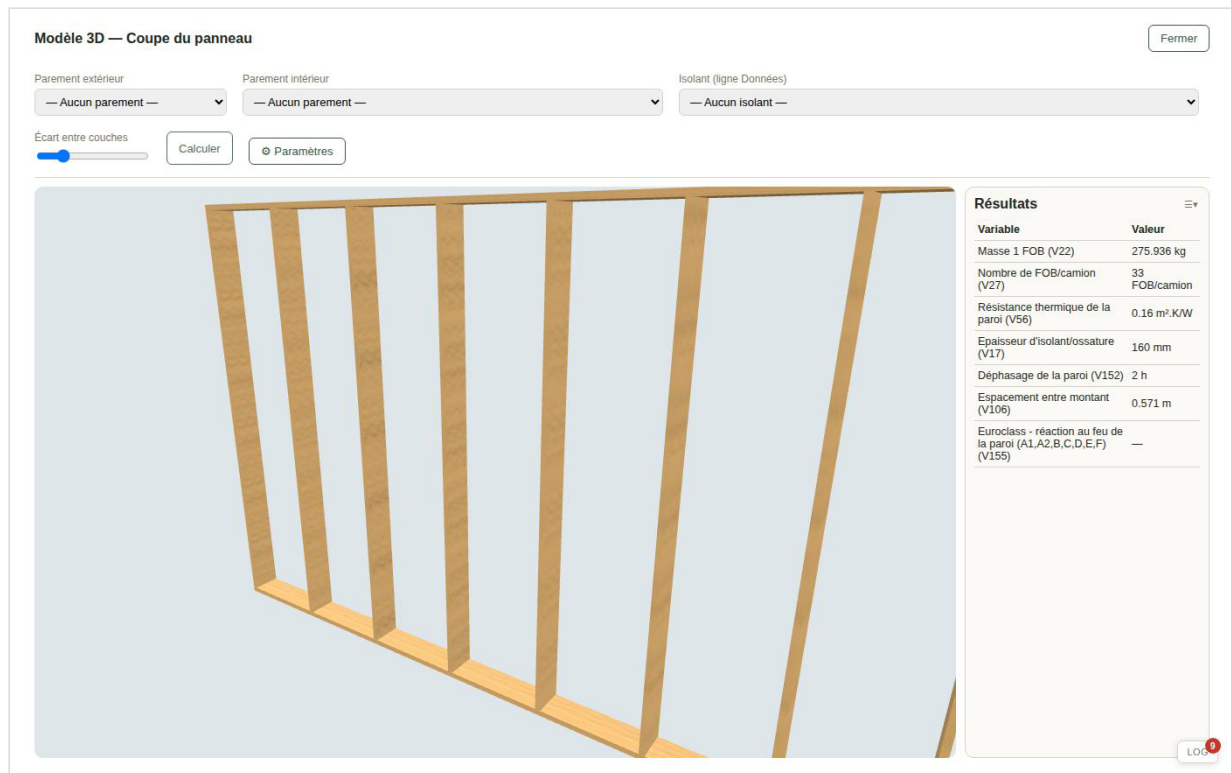


Figure 15. After calculation: the selected results (wall thermal resistance, phase shift, mass, cost...) are displayed on the right, updated according to the selections made in the 3D Model.

The 3D Model replaces the Iterative menu's calculation while it is open: its "Calculate" button becomes the reference for the whole project while it is in use.